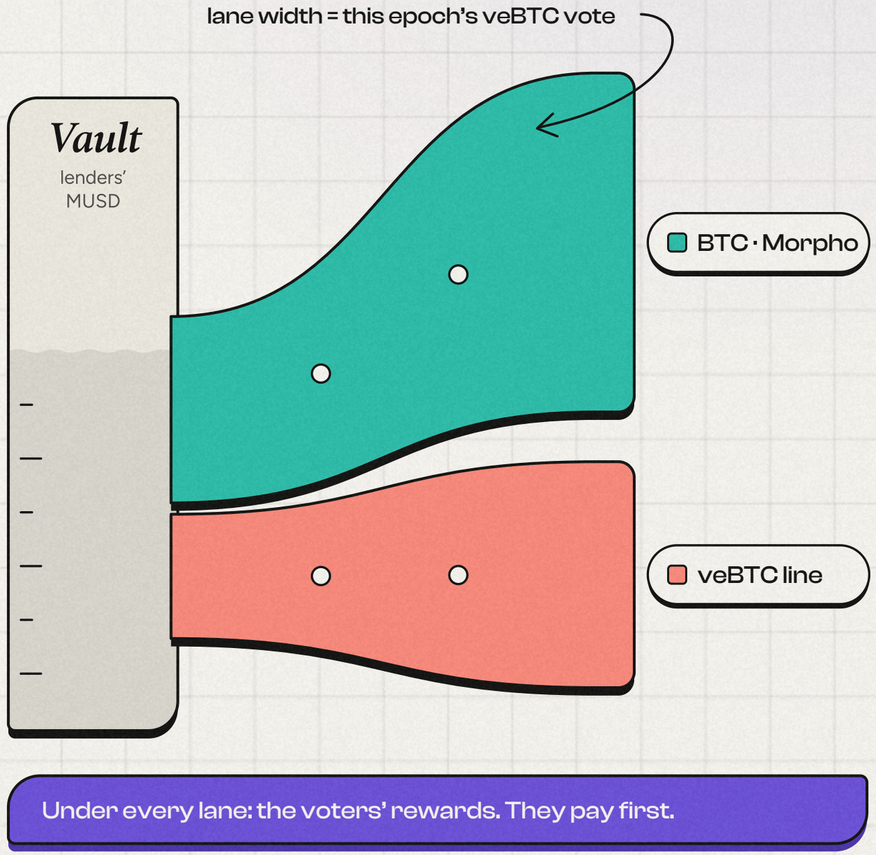
Surety

Lending gauges for Mezo

veBTC holders decide where MUSD credit goes, and their rewards take *the first loss*.

Lend MUSD with the voters' rewards standing in front of you, or borrow against your veBTC without unlocking it.



Live on Mezo mainnet · v2

surety-credit.vercel.app

Yehor Ivashchenko

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The problem

Plenty of BTC. Almost no *credit*.

804 BTC

locked in veBTC

Held by 1,805 holders. None of it can be borrowed against without unlocking.

0

loss for whoever picks

In money markets a curator or governance decides where deposits go, and carries none of the loss when the pick goes bad.

-83%

MUSD in sMUSD since 5 Nov

Mezo's savings vault fell from 7.38M to 1.29M MUSD, and its yield comes in lumps. MUSD holders need somewhere productive and safe to earn.

THIS WEEK ON MEZO

5 Nov: an Enclave repaid 10.51M MUSD and pulled its liquidity.

MUSD supply `29.48M → 19.02M` sMUSD `7.38M → 1.29M` MUSD in the CL mUSDC/MUSD pool `5.33M → 7.5k`

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From Mezo's blog · "Aerodrome for lending"

"...the system cannot tell who is serious"

mezo.org/blog/aerodrome-for-lending

WHAT MEZO ASKED FOR

Lending gauges: veBTC votes steer capital to lending markets, as a risk signal.

WHAT IT LEFT OPEN

The mechanism. It is saved for "a future post".

SURETY'S ANSWER

Make the vote cost something. The voters' rewards take the first loss.

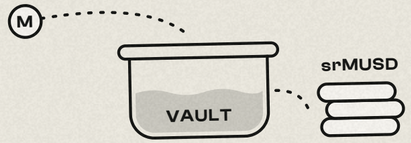
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How it works

Four verbs, *one* vault.

Lenders bring the MUSD. veBTC holders steer it and stand behind their picks. Borrowers draw it with no redemption risk.

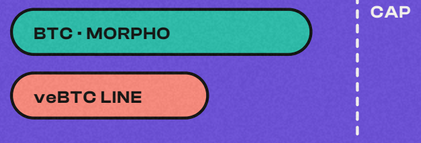
01 for lenders



Lend.

Deposit MUSD into a Morpho VaultV2. You hold srMUSD; idle MUSD waits in Mezo's sMUSD.

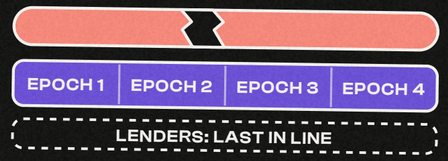
02 for veBTC holders



Vote.

veBTC power is snapshotted at epoch start. The allocator moves capital to match the votes, inside vault caps.

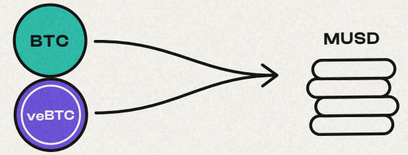
03 for the voters, again



Underwrite.

Voters earn 10% of interest plus sponsor incentives. Those rewards season 4 epochs and pay first if a line they backed goes bad.

04 for borrowers



Borrow.

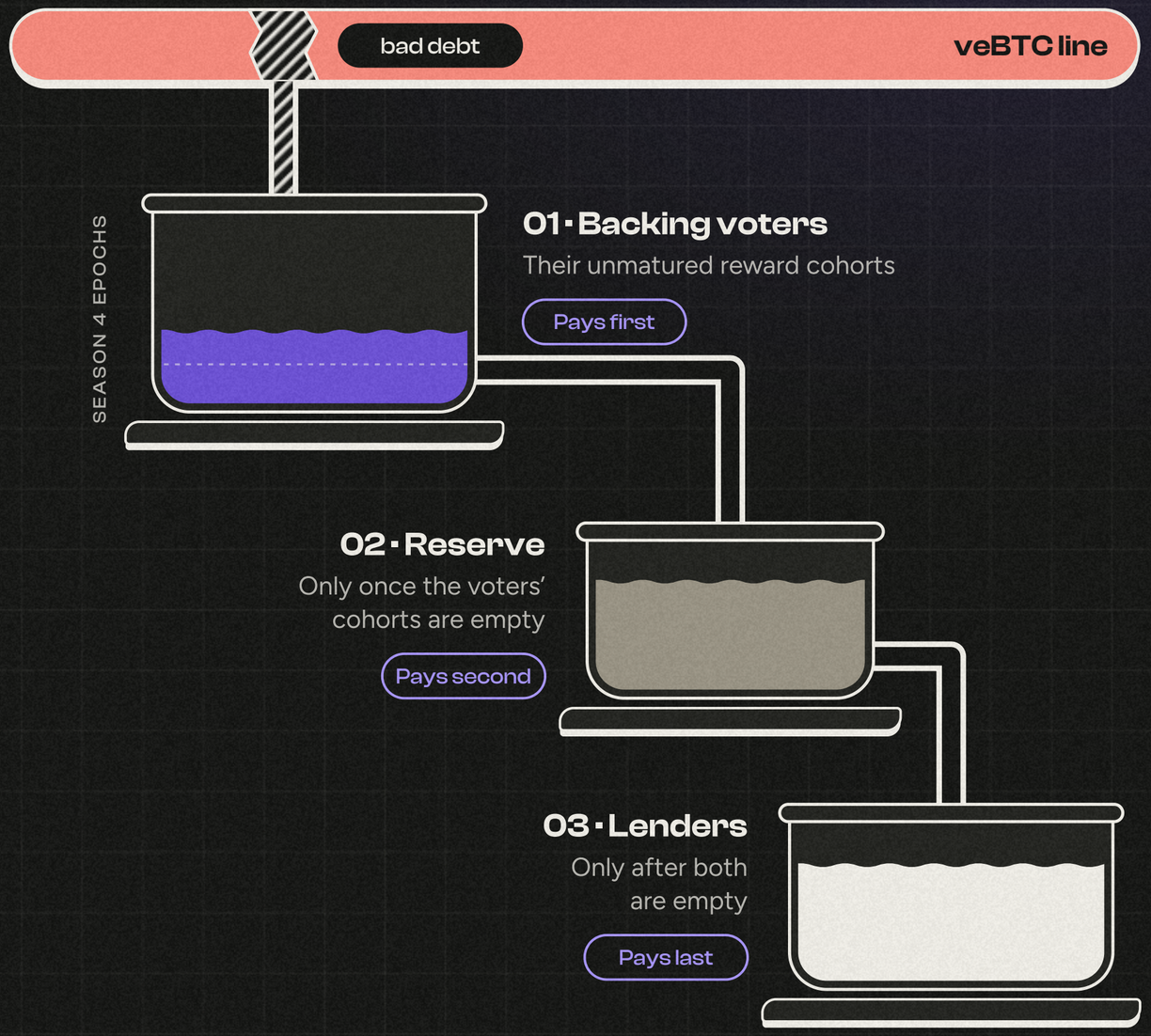
MUSD with no redemption risk: against BTC on Mezo's Morpho Blue, or against a locked veBTC that keeps voting.

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Who pays first

When a loan goes bad, the voters pay first.

Order	Backing voters’ unmatured cohorts, then the reserve, then lenders.
Size	Cover is sized so lenders return to the pre-loss share price.
Gate	New deposits pause while a cover is re-recognised, so nobody can skim it.



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What changed since Wave 1

Seven review points. Six *done*.

	THE WAVE 1 REVIEW ASKED FOR	WHAT V2 SHIPPED	STATUS ON 12 NOV
01	**Escrow vote and claims on chain**	Bribe, fee and rebase claims on testnet; a harvest paid down the loan	Done
02	**Real MUSD in the mainnet vault**	2.5 MUSD in, lent on Mezo's own Morpho Blue and against veBTC	Done
03	**Outside users borrowing**	Starter kits, a no-install Try it and a public users ledger	No outside users yet
04	**A MUSD/USD price, not $1**	MusdPeg reads Mezo's pools; a borrow at $0.94 reverted on chain	Done
05	**A wider trigger-to-floor gap**	Auctions start at 65% LTV and stop at an 85% floor (was 70% and 70%)	Done
06	**Who pays the team**	Four protocol fees to a treasury, each one paid on chain	Done
07	**The app loads fast**	Static first paint, time-boxed RPCs, checked in headless Chrome	Done

Every point with its commits, transactions and checks: `FEEDBACK-RESPONSE.md` in the repository

The MUSD peg

Priced by Mezo's *own* pools.

Sources	Four of Mezo's pools: CL and basic mUSDC/MUSD, basic MUSD/mUSDT, and CL BTC/MUSD × Skip BTC/USD.
Checks	A CL source needs a liquidity floor over its 30-minute window; a basic pool, an observation under 6 h old.
Median	Healthy when 2 or more sources read, agree within 1,000 bps, and the median is within 300 bps of $1.
Clamp	BTC in MUSD is BTC/USD ÷ the median, clamped to [$1, $1.03]. Never above BTC/USD.
Breaker	While unhealthy, new borrowing and new allocations stop. Repayments, auctions and withdrawals go on.

MEZO MAINNET · AT LAUNCH — block 13,118,945

~~CL mUSDC/MUSD~~	*under its liquidity floor*
Basic mUSDC/MUSD	0.991271
~~Basic MUSD/mUSDT~~	*13.8 h old, past 6 h*
CL BTC/MUSD × Skip	0.999005

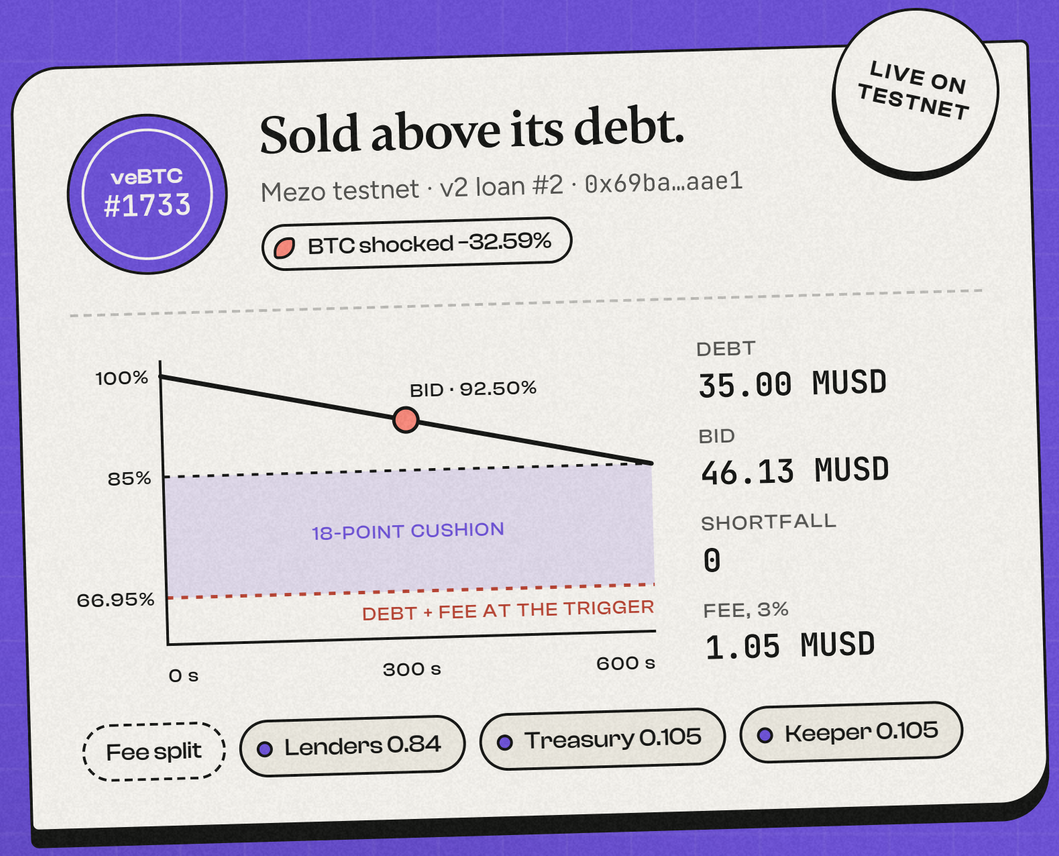
0.9951

Healthy

2 of 4 sources read, 78 bps apart and 49 bps under $1: borrowing open, BTC priced at BTC/USD.

TESTNET MUSD forced to $0.94. A borrow sent anyway reverted on chain with **`PegUnhealthy`**. `0x6438…8e16`

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In v1 an auction could clear below the debt by design. In v2 the floor sits above it.

Liquidations

An 18-point cushion before any shortfall.

Trigger	65% LTV, or lock expiry. Was 70%.
Floor	The auction falls from 100% of the lock's value to an 85% floor. Was 70%.
Cushion	At the trigger, debt plus the 3% fee is at most 66.95%. The contract refuses a floor below that.
Fee split	80% to lenders, 10% to the treasury, 10% to whoever starts the auction.

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For veBTC holders

Your veBTC repays its *own* loan.

01 for the borrower

Lock.

Borrow up to 50% LTV, 40% on a permanent lock. The NFT moves into its own escrow.

02 on Mezo

Vote.

The escrow still votes the NFT on Mezo's gauges, wherever the borrower picks.

03 on Mezo

Earn.

Fees and bribes accrue to the NFT. The veBTC rebase grows the lock.

04 anyone can call it

Repay.

With auto-repay on, a harvest turns the MUSD into repayment. Other tokens go to the borrower.

On chain · Mezo testnet · v2 loan #1, veBTC #1732

AUTO-REPAY	ESCROW VOTE	BRIBE	HARVEST · 12 NOV
Switched on	100% on MUSD/mUSDC	20 MUSD on that gauge	0.1676 MUSD repaid
0x30a8…fc41	0x1a50…0430	0x203f…ad0f	0x04be…09e6

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For lenders

Idle cash in sMUSD. A market for *lender shares*.

sMUSD idle cash, parked on Mezo

The vault's idle MUSD, its 10% buffer included, sits in Mezo's sMUSD as VaultV2's liquidity adapter. Withdrawals pull from it.

100 MUSD deposit lands in sMUSD
`0xe25a…51f8`

Rebalance sends 108 to the lines; 12 stays parked
`0x5fe4…c5b2`

Honest yield. sMUSD pays in lumps: near zero from 23 Oct to 9 Nov, then about 5.6% a year on our 2.5 MUSD over 10–12 Nov. It keeps cash productive on Mezo; we don't count it as a return.

ShareMarket a way out when the cash is lent

When the vault's cash is lent out and a redeem would fail, a lender can sell srMUSD at a discount to the live share price.

20 srMUSD listed 1% under the share price
`0x7021…af2f`

10 srMUSD filled for 9.9004 MUSD; 0.0099 to the treasury
`0x741e…de8a`

No owner. Fills wait while deposits are paused after a loss; `cancelAndRedeem` stays open.

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Business

Small fees. Break-even near *4.1M*.

Borrowers BTC and veBTC

Borrow interest, origination and liquidation fees

LENDERS

Interest

The rest of the borrow interest, plus 80% of every liquidation fee.

veBTC VOTERS

10%

Of interest, for standing first in line for losses. Plus sponsor incentives.

TREASURY

4 fees

`0.25%` origination
`0.20%`/yr management
`10%` of liquidation fees
`0.10%` of share sales

TREASURY INCOME · MUSD A YEAR

At the caps 4,000 MUSD in the vault	20.5
At $1M	5,118
At $10M	51,180
Running costs operations and one review a year	-21,200

Fees cover the costs from about 4.1M MUSD in the vault.

WHO PAYS THE TEAM

Nobody yet.

Surety

Engineering & security

Tested like it already holds *money*.

379

Foundry tests

Including 15 invariants at 256 runs × 64 actions, and 16 review regressions.

173

TypeScript tests

Across the SDK, the keeper, the MCP server and the web app.

0

open high or medium

Across two independent reviews, of v1 and of the v2 changes.

19/19

exact match on Sourcify

Every v2 mainnet contract. Mezo's Blockscout lists no compilers right now.

v2 review	`1 high · 2 medium · 2 low`	All five fixed, each with regression tests
v1 review	`4 medium · 6 low`	All four medium and 5 of 6 low fixed; 1 low acknowledged

Not an external audit yet. Owner roles stay on one key until a timelock and a multisig take over.

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Proof

Deployed, verified, *run* on chain.

v2 · Mezo mainnet

deployed 9 Nov

MARKET	`0x4bff245d…3030b8` on Mezo's own Morpho Blue, LLTV 86%
CONTRACTS	19, all exact match on Sourcify
LIMITS	2,000 MUSD cap per line, 1-day timelocks
PEG	0.9951 at launch, healthy
REAL MUSD	2.5 in: 1.125 on Mezo's Morpho Blue, 1.125 in the veBTC line, 0.25 in sMUSD
LOANS	1 MUSD borrowed on Morpho; 1 MUSD against veBTC, its escrow voting on Mezo

v2 · Mezo testnet

53 transactions

The whole lifecycle, each role from its own account, on Mezo's testnet Morpho Blue.

- Deposit into sMUSD
- veBTC loans, fees paid
- `PegUnhealthy` revert
- ShareMarket fill
- Escrow claims
- Vote and rebalance
- Escrow vote and bribe
- Auction, no shortfall
- Starter kit
- Auto-repay harvest

every transaction: `/app/proof`

v1 · kept

TESTNET LOSS RUN

157.16 MUSD of bad debt; the voters' rewards covered 120.18 (76%)

MAINNET

11 contracts, capped, verified

TESTNET

14 contracts, verified

CHECK IT

`surety-credit.vercel.app/app/review`

`surety-credit.vercel.app/app/proof`

`github.com/RaYYeR220/surety`

Surety

Try it

A testnet loan in about a minute.

No wallet and nothing to install. A key kept in the browser signs, and a starter kit pays for everything.

`surety-credit.vercel.app/app/try`

No time? `/app?demo=1` shows every page on example data.

`/app/ledger` Every wallet that has used Surety, on both networks. Team wallets are labelled, not hidden.

The path

Mezo testnet

01 **Claim a starter kit**
0.00005 BTC for gas, a 0.001 BTC veBTC lock and 30 test MUSD. One per wallet.

02 **Borrow against the kit's veBTC**
The lock moves into an escrow of its own and stays yours.

03 **Turn on auto-repay**
Anyone may then harvest its rewards into repayment.

04 **Vote the escrowed veBTC on Mezo**
The escrow votes Mezo's pools while the loan is open.

05 **Repay**
The loan closes and the veBTC comes back.

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Roadmap · team

Capped first. Then *open it up*.

NEXT

A Mezo vault gauge for lenders, through governance

`ERC4626VaultAdapter`

Owner roles behind a timelock and a multisig

BEFORE CAPS RISE

An external audit

THEN

Raise the caps

More BTC collateral lines

BUILT BY

Yehor Ivashchenko solo

X

`x.com/BazydloMis57623`

APP

`surety-credit.vercel.app`

CODE

`github.com/RaYYeR220/surety`

VIDEO

`youtu.be/6GFWaNuHvXo`